

CS471 – Block 3 Worksheet Solutions

Lessons 18-23: Probability, Bayesian Networks, and Temporal Models

Block 3 Equations Reference

- **Joint Probability (Independent Events):** *Computes the probability of both A and B occurring when the outcome of one does not mathematically affect the other.*

$$P(A \cap B) = P(A)P(B)$$

- **Law of Total Probability (Marginalization):** *Computes the total marginal probability of event B by summing the probabilities of B occurring across all mutually exclusive conditions of prior event A .*

$$P(B) = P(B|A)P(A) + P(B|\neg A)P(\neg A)$$

- **Bayes' Rule:** *Updates the probability of hypothesis A given new evidence B , using the likelihood $P(B|A)$, prior $P(A)$, and marginal evidence $P(B)$.*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- **Bayesian Network Factorization:** *Calculates the full joint probability of a network by multiplying the conditional probability of each node given only its direct parents, exploiting conditional independence.*

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$$

- **Markov 2-Step Transition:** *Computes the probability of reaching a state in two steps by summing over all possible intermediate transition states x_1 .*

$$P(X_2 \mid X_0) = \sum_{x_1} P(X_2 \mid x_1)P(x_1 \mid X_0)$$

- **HMM Filtering Update:** *Updates the belief of the current hidden state X_t by projecting the previous belief forward (Prediction) and multiplying by the new sensor likelihood (Update), scaled by normalization constant α .*

$$B(X_t) = \alpha P(e_t|X_t) \sum_{x_{t-1}} P(X_t|x_{t-1})B(x_{t-1})$$

Part I: Guided Practice

Lesson 18: Quantifying Uncertainty

An autonomous UAV relies on two critical payloads for an intelligence mission: a GPS navigation module and a tactical Datalink. The probability that the GPS module is operational is $P(G) = 0.88$, and the probability that the Datalink is operational is $P(D) = 0.95$. Assume the two subsystem events are independent.

- (a) Compute the probability that both subsystems are operational.

Solution

$$\begin{aligned} P(G \cap D) &= P(G)P(D) \\ &= (0.88)(0.95) \\ &= \mathbf{0.836} \end{aligned}$$

- (b) Given that the GPS module is operational, compute the probability that the Datalink is operational.

Solution

$$\begin{aligned} P(D|G) &= P(D) \\ &= \mathbf{0.95} \end{aligned}$$

This holds because G and D are independent.

- (c) Explain how independence affects the calculations in parts (a) and (b). State what additional information would be needed if the subsystem events were dependent.

Solution

Independence allows the joint probability to be computed by multiplying the marginal probabilities, and conditioning on one event does not change the probability of the other. If the events were dependent, a conditional probability such as $P(D|G)$ or equivalent joint-probability information would be required.

Lesson 19: Belief Update (Bayes' Rule)

A military network uses an automated intrusion detection system (IDS) to identify Advanced Persistent Threat (APT) activity. Historical data indicate that 4% of networks experience an APT intrusion during a deployment. If an APT is present, the IDS reports a critical alert 92% of the time. If no APT is present, the IDS produces a false alert 6% of the time. Let A represent an APT presence and L represent an IDS alert.

- (a) Compute the probability that the IDS reports an alert, $P(L)$.

Solution

$$\begin{aligned}P(L) &= P(L|A)P(A) + P(L|\neg A)P(\neg A) \\&= (0.92)(0.04) + (0.06)(0.96) \\&= 0.0368 + 0.0576 \\&= \mathbf{0.0944}\end{aligned}$$

- (b) Use Bayes' Rule to compute the probability that an APT is present given that the IDS reports an alert, $P(A|L)$.

Solution

$$\begin{aligned}P(A|L) &= \frac{P(L|A)P(A)}{P(L)} \\&= \frac{(0.92)(0.04)}{0.0944} \\&= \frac{0.0368}{0.0944} \\&\approx \mathbf{0.3898} \text{ (or 38.98\%)}\end{aligned}$$

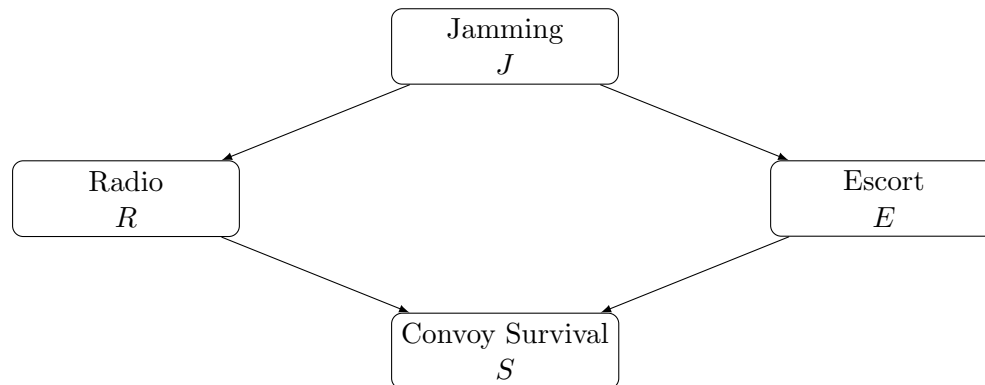
- (c) Explain why $P(A|L)$ is substantially lower than 92%, even though the IDS identifies 92% of actual APT intrusions.

Solution

Actual APT intrusions are rare (4% base rate), so the large volume of normal network traffic creates many opportunities for false alerts. The 92% value is the likelihood $P(L|A)$, not the posterior probability $P(A|L)$.

Lesson 20: Dependency Modeling and Factorization

A logistics convoy planning system uses the Bayesian network below. Enemy Jamming (J) affects both Radio Communications (R) and Drone Escort capability (E). Radio Communications and Drone Escorts then directly affect Convoy Survival (S).



- (a) Identify the parent or parents of R , E , and S .

Solution

The parent of R is J . The parent of E is J . The parents of S are R and E .

- (b) Identify the child or children of J , R , and E .

Solution

The children of J are R and E . The child of R is S . The child of E is S .

- (c) State one conditional independence relationship implied by the network and briefly explain why it holds.

Solution

$R \perp E \mid J$. Once the enemy jamming environment is known, learning the radio status provides no additional information about the drone escort capability under the assumptions of this network. (Another valid relationship is $J \perp S \mid R, E$).

- (d) Factor the full joint probability $P(J, R, E, S)$ using the Bayesian network.

Solution

$$P(J, R, E, S) = P(J)P(R|J)P(E|J)P(S|R, E)$$

- (e) List the probability distribution or conditional probability table associated with each node. Briefly explain why this factorization can require fewer probability values than the full joint distribution.

Solution

The required distributions are $P(J)$, $P(R|J)$, $P(E|J)$, and $P(S|R, E)$. The full joint distribution assigns probabilities to every possible combination of all variables. The Bayesian network instead represents each variable using only its parents, encoding conditional independence assumptions that reduce the number of required probability values.

Lesson 21: Probabilistic Inference

Using the same convoy network, assume reliable Radio Communications (R) have already been observed by the command post. Previous analysis determined that $P(E|R) = 0.70$. The remaining probabilities for survival are $P(S|E, R) = 0.96$ and $P(S|\neg E, R) = 0.40$.

- (a) Compute the probability of convoy survival given reliable radio communications, $P(S|R)$.

Solution

$$\begin{aligned} P(S|R) &= P(S|E, R)P(E|R) + P(S|\neg E, R)P(\neg E|R) \\ &= (0.96)(0.70) + (0.40)(0.30) \\ &= 0.672 + 0.120 \\ &= \mathbf{0.792} \end{aligned}$$

- (b) Interpret your answer in the mission context.

Solution

Given reliable radio communications but uncertain drone escort status, the probability of convoy survival is 79.2%. The result mathematically combines both possible drone escort conditions according to their conditional probabilities.

Lesson 22: Temporal Prediction (Markov Models)

An Electronic Warfare officer models the operational mode of an adversary's radar using two states: T = Tracking, S = Searching. The one-step transition matrix is:

$$P = \begin{array}{c|cc} & T & S \\ \hline T & 0.75 & 0.25 \\ S & 0.40 & 0.60 \end{array}$$

The radar is currently in Tracking mode.

- (a) What is the probability that the radar is in Searching mode after one time step?

Solution

$$P(X_1 = S \mid X_0 = T) = \mathbf{0.25}$$

This follows directly from the transition matrix since the radar is currently in Tracking mode.

- (b) Compute the probability that the radar is in Searching mode after two time steps.

Solution

The radar can be in Search after two steps through either path $T \rightarrow T \rightarrow S$ or $T \rightarrow S \rightarrow S$.

$$\begin{aligned} P(X_2 = S \mid X_0 = T) &= P(T|T)P(S|T) + P(S|T)P(S|S) \\ &= (0.75)(0.25) + (0.25)(0.60) \\ &= 0.1875 + 0.1500 \\ &= \mathbf{0.3375} \text{ (or } 33.75\%) \end{aligned}$$

- (c) State the Markov assumption and explain how it applies to this model.

Solution

The Markov assumption states that the probability distribution of the next state depends only on the current state, not on the complete sequence of earlier states. In this model, once the radar's current mode is known, its earlier operational history is not used to determine the next transition.

Lesson 23: Hidden State Reasoning (HMMs)

A naval sonar operator cannot directly observe the propulsion state of a hostile submarine: Quiet (Q) or Cavitating (C). An acoustic buoy registers a Detection (D) providing imperfect evidence. Because the submarine is currently Quiet, the transition model gives $P(Q) = 0.70$ and $P(C) = 0.30$ for the next step. Observation probabilities are $P(D|Q) = 0.10$ and $P(D|C) = 0.85$.

- (a) Identify the hidden state and the observable evidence. Explain why the propulsion state is hidden.

Solution

The hidden state is the submarine's actual propulsion mode (Quiet or Cavitating). The observable evidence is whether a Detection is registered on the sonobuoy. The condition is hidden because it cannot be observed directly by the operator; they must infer it from imperfect acoustic evidence.

- (b) Given a Detection is registered, compute the probability that the submarine is Cavitating at the next time step, $P(C|D)$.

Solution

First, compute the marginal probability of a detection:

$$\begin{aligned} P(D) &= P(D|C)P(C) + P(D|Q)P(Q) \\ &= (0.85)(0.30) + (0.10)(0.70) \\ &= 0.255 + 0.070 \\ &= 0.325 \end{aligned}$$

Now, apply Bayes' Rule:

$$\begin{aligned} P(C|D) &= \frac{P(D|C)P(C)}{P(D)} \\ &= \frac{(0.85)(0.30)}{0.325} \\ &= \frac{0.255}{0.325} \\ &\approx \mathbf{0.7846} \end{aligned}$$

- (c) Briefly interpret the result.

Solution

After receiving a buoy detection, the estimated probability that the submarine is cavitating is approximately 78.5%. The acoustic alert heavily increases belief in the cavitating state, but it does not prove it with 100% certainty because quiet submarines can also generate anomalous detections.

Part II: Independent Student Practice

Practice 1: Quantifying Uncertainty

A cyber defense node uses two distinct security measures: a Hardware Firewall (F) and a Software Antivirus (A). $P(F) = 0.90$ and $P(A) = 0.80$. Assume these are independent.

- (a) Compute the probability that both security measures are operational, $P(F \cap A)$.

Solution

$$\begin{aligned} P(F \cap A) &= P(F)P(A) \\ &= (0.90)(0.80) \\ &= \mathbf{0.72} \end{aligned}$$

- (b) Compute $P(A|F)$ and explain why this value makes sense structurally.

Solution

$$P(A|F) = \mathbf{0.80}$$

Because the events are independent, knowing the firewall is operational provides zero mathematical information about the antivirus state.

Practice 2: Belief Update (Bayes' Rule)

An automated drone maintenance AI checks for Engine Stress (E). Historical logs show an engine stress base rate of 5% ($P(E) = 0.05$). The onboard diagnostic sensor triggers a Warning (W) 95% of the time when stress is present. It triggers a false warning 10% of the time on healthy engines.

- (a) Calculate the marginal probability of a warning, $P(W)$.

Solution

$$\begin{aligned} P(W) &= P(W|E)P(E) + P(W|\neg E)P(\neg E) \\ &= (0.95)(0.05) + (0.10)(0.95) \\ &= 0.0475 + 0.0950 \\ &= \mathbf{0.1425} \end{aligned}$$

- (b) Calculate the posterior probability of actual engine stress given a warning, $P(E|W)$.

Solution

$$\begin{aligned}
 P(E|W) &= \frac{P(W|E)P(E)}{P(W)} \\
 &= \frac{0.0475}{0.1425} \\
 &\approx \mathbf{0.3333} \text{ (33.33\%)}
 \end{aligned}$$

Practice 3 & 4: Factorization and Inference

A satellite constellation is modeled with the following network: Solar Flare (F) affects both the Comm Link (C) and the Power Grid (P). Both C and P directly affect Mission Success (S).

- (a) Draw the Bayesian network graph and write out the full factored joint probability $P(F, C, P, S)$.

Solution

Graph: $F \rightarrow C, F \rightarrow P, C \rightarrow S, P \rightarrow S$.

Factorization:

$$P(F, C, P, S) = P(F)P(C|F)P(P|F)P(S|C, P)$$

- (b) Assume the Comm Link (C) has been observed as fully operational. We know $P(P|C) = 0.85$. The survival probabilities are $P(S|P, C) = 0.98$ and $P(S|\neg P, C) = 0.45$. Calculate $P(S|C)$.

Solution

$$\begin{aligned}
 P(S|C) &= P(S|P, C)P(P|C) + P(S|\neg P, C)P(\neg P|C) \\
 &= (0.98)(0.85) + (0.45)(0.15) \\
 &= 0.833 + 0.0675 \\
 &= \mathbf{0.9005}
 \end{aligned}$$

Practice 5: Temporal Prediction (Markov Models)

A cyber analyst models a server node as either Secure (C) or Compromised (K). The transition matrix is:

$$P = \begin{array}{c|cc} & C & K \\ \hline C & 0.90 & 0.10 \\ K & 0.60 & 0.40 \end{array}$$

The server is currently Secure. Compute the probability it will be Compromised after **two** time steps.

Solution

The server is initially Secure (C). To be Compromised (K) at $t = 2$, it must transition through paths $C \rightarrow C \rightarrow K$ or $C \rightarrow K \rightarrow K$.

$$\begin{aligned}P(X_2 = K \mid X_0 = C) &= P(C|C)P(K|C) + P(K|C)P(K|K) \\&= (0.90)(0.10) + (0.10)(0.40) \\&= 0.09 + 0.04 \\&= \mathbf{0.13} \text{ (13\%)}\end{aligned}$$

Practice 6: Hidden State Reasoning (HMMs)

A UAV seeks a ground target that is either Present (T) or Absent (A). The UAV's radar yields a Ping (P). Because the target is currently Absent, the transition model for the next step is $P(T) = 0.15$ and $P(A) = 0.85$. The radar emission likelihoods are $P(P|T) = 0.90$ and $P(P|A) = 0.05$. If a Ping is observed on the next step, what is the updated probability that the target is Present, $P(T|P)$?

Solution

First, compute the marginal probability of a Ping:

$$\begin{aligned}P(P) &= P(P|T)P(T) + P(P|A)P(A) \\&= (0.90)(0.15) + (0.05)(0.85) \\&= 0.135 + 0.0425 \\&= 0.1775\end{aligned}$$

Update belief using Bayes' Rule:

$$\begin{aligned}P(T|P) &= \frac{P(P|T)P(T)}{P(P)} \\&= \frac{0.135}{0.1775} \\&\approx \mathbf{0.7606} \text{ (76.06\%)}\end{aligned}$$